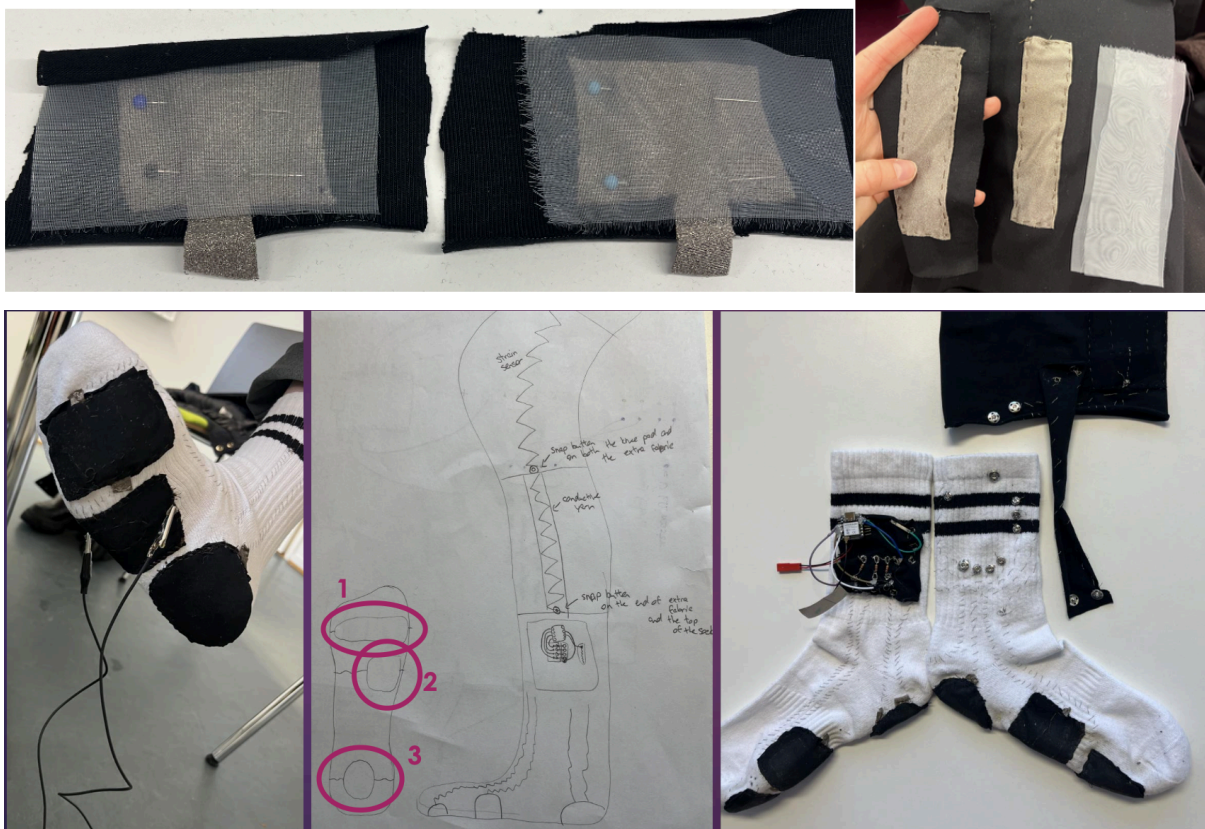


Smart socks - Smart Wearables ELEC-E7840 course spring 2026

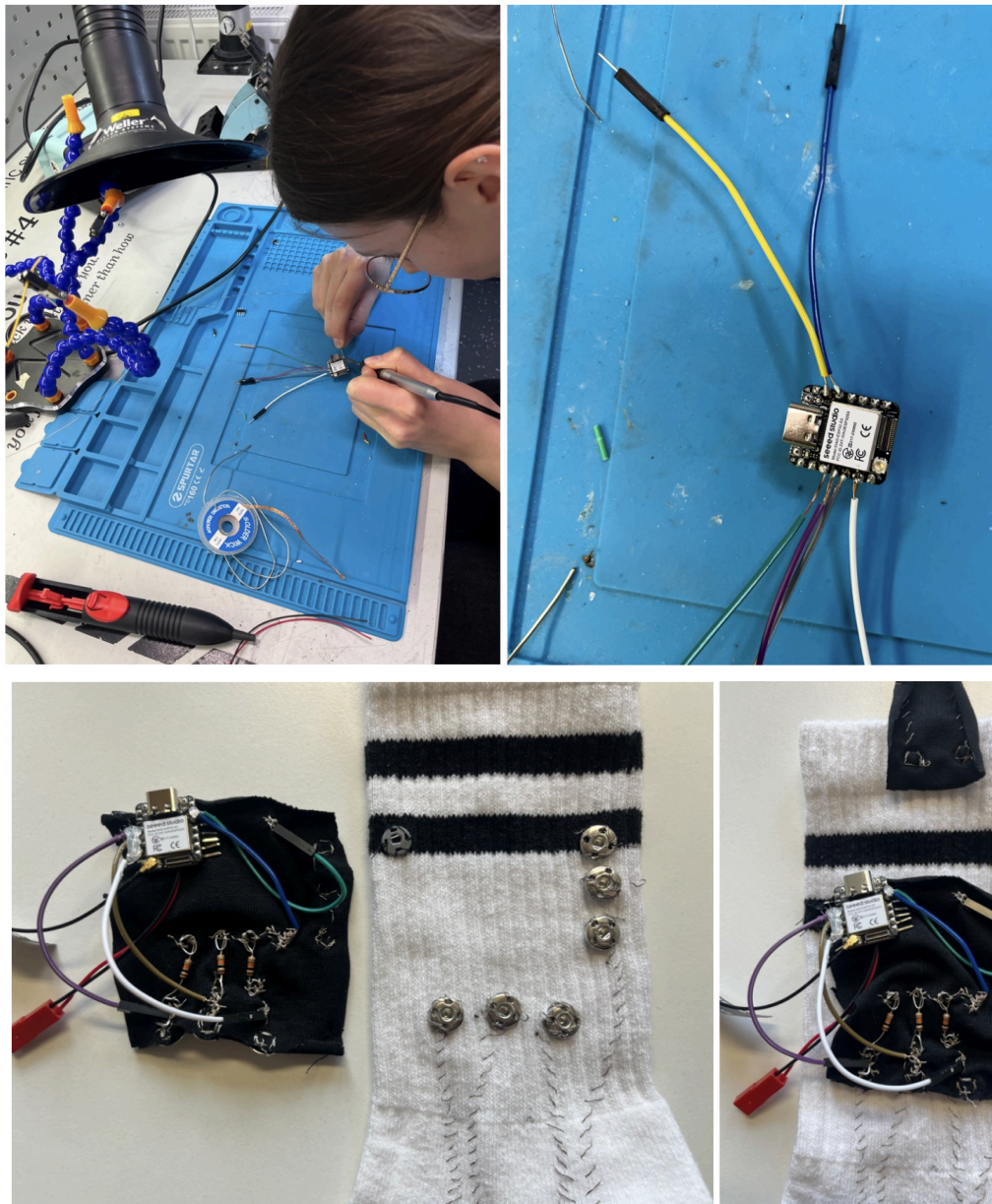
Together with my partner, I designed and implemented a functional wireless smart clothing prototype (smart socks and a knee brace) for real-time physical activity recognition. My specific responsibilities included integrating the hardware into the socks, Arduino programming, and building a machine learning model for real-time recognition.

The project was roughly divided into the following phases:

1. Smart textile manufacturing (E-textiles): I fabricated soft piezoresistive pressure sensors using a multi-layer textile structure, conductive fabrics, and conductive threads. I positioned the sensors at critical pressure points on the sole of the foot (heel, ball of the foot, and arch) as well as on the knee brace.



2. Embedded electronics and circuitry: I designed and soldered voltage divider circuits without a breadboard and integrated them directly into the socks using snap fasteners to ensure washability and wearability. I programmed two Seeed XIAO ESP32S3 microcontrollers to read sensor data and transmit it wirelessly via Wi-Fi.



3. Signal processing and machine learning: I built a Python tool for real-time data reception, preprocessing, and feature extraction (sliding time window, statistical features, normalization). I trained and optimized a classification model (Random Forest) capable of recognizing eight different activities and postures (e.g., walking, using stairs, sit-ups, cross-legged sitting) with 95% validation accuracy.

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report, confusion_matrix
from sklearn.preprocessing import MinMaxScaler
import joblib

import pandas as pd

df_master = pd.read_csv('final_training_data.csv')
sensor_cols = ['Right_HEEL', 'Right_TOES', 'Right_ARCH', 'Right_KNEE', 'Left_HEEL', 'Left_Arch', 'Left_Toes']
target_names = ['walking', 'Stairs_down', 'Sitting', 'Standing', 'Sit-to-Stand', 'Cross-legged', 'Stairs_up', 'Unknown']
scaler = MinMaxScaler()
df_master[sensor_cols] = scaler.fit_transform(df_master[sensor_cols])
print(df_master.head())

   Right_HEEL  Right_TOES  Right_ARCH  Right_KNEE  Left_HEEL  Left_Arch \
0    0.996083    0.988229    0.986596    0.368112    0.982926    0.987523
1    0.995349    0.973203    0.979767    0.325986    0.988002    0.987523
2    0.867564    0.207363    0.194740    0.341661    0.994924    0.993993
3    0.995104    0.228400    0.236216    0.505021    0.994924    0.993993
4    0.994370    0.887553    0.886697    0.210629    0.991232    0.992606

   Left_Toes  Activity_Name  Label
0    0.994513      walking      0
1    0.996982      walking      0
2    0.996982      walking      0
3    0.995062      walking      0
4    0.996708      walking      0

def extract_window_features(df, window_size=30, step_size=10):
    X = []
    y = []

    for activity in df['Activity_Name'].unique():
        temp_df = df[df['Activity_Name'] == activity].copy()

        for i in range(0, len(temp_df) - window_size, step_size):

```

Screenshot of one of the codes used for the project.

4. User testing and evaluation: We collected and analyzed data from five external test subjects and measured the system's usability and comfort using a structured questionnaire.

What did this course and project teach me?

- I gained practical insight into how piezoresistive materials respond to mechanical pressure and compression, and how using different material layers prevents sensor saturation under extreme pressure.
- I deepened my knowledge of electronics fundamentals, circuit design, and the use of conductive thread. After a hiatus of nearly ten years, I also had the opportunity to visit an electronics workshop, as we needed to solder wires to an ESP32 microcontroller to make the system wireless.
- I learned to write C++/Arduino code for the microcontroller to collect analog data and to wirelessly synchronize data transmission from two separate devices to a computer.

- I learned to execute the entire machine learning pipeline, all the way from data cleaning and normalization to comparing classifier performance (Decision Tree vs. Random Forest) and analyzing confusion matrices. The most important acknowledgement was utilizing an "Unknown" class to improve model reliability.

My key insights

- The circuit works in theory, but in the context of a garment, it stretches, twists, and is subjected to constant mechanical stress. My most significant insight was to move the microcontroller and rigid connections off the flexible sock and onto separate panels attached via snaps, while reinforcing the solder joints with a non-conductive material (hot glue). Ensuring a garment is wearable and durable requires just as much mechanical design as it does electronics. This also highlighted the fact that design and testing go hand in hand: without testing, a design can be completely unusable.
- Machine learning quality depends on physical fit and not just code: The challenges encountered with the machine learning model when used by different people (accuracy ranging from 65% to 87%) stemmed not from the algorithm itself, but from variations in foot size and gait. These differences caused the sock to stretch in various ways, leading to anomalies in the underlying signal. I realized how critical user-centric design, sensor adjustability, and data diversity are for the generalizability of smart clothing.
- Iterative "trial and error" and tolerating uncertainty: As I had never worked with circuitry or wearable electronics before, the project taught me perseverance. Dealing with broken solder joints, debugging code, and facing multiple model failures right before the live demo was frustrating, but we were able to recover from that. All together our live demo and written overview from the project gave us the highest grade (5/5).

Results

Key Findings:

- ▶ **Static activities (sitting, standing):** ~90–100%
- ▶ **Walking:** generally high accuracy (>85%)
- ▶ **Stairs & transitions:** most challenging
 - ▶ Similar patterns to walking and high variability between users



Figure 9. Results from comparing the two models. Other similar plots are found in the code zip files for all the five test users. Plots by Aino Karjalainen.

Results

MODEL PERFORMANCE

MODEL	TRAINING ACCURACY	KEY OBSERVATION
Decision tree	72 %	Poor on dynamic activities
Random forest	95 %	Better performance, improvement with more data

Accuracy in random forest increased from **90% to 95%** after the live demo
 → due to **adding more training data**, improving model generalization.

TEST USERS (GENERALIZATION, N=5):

USER	ACCURACY	NOTES
1	70 %	Issues with stairs & sitting
2	79%	Most stable performance
3	87 %	High accuracy, some bias initially, dynamic movements have variation
4	65 %	Confusion in standing
5	80 %	Strong overall, weak in stairs down

Why results look like this:

- Static postures → **stable and distinct sensor signals**
- Dynamic movements → **overlapping patterns + noise**
- Performance strongly depends on:
 - Training data amount
 - **User differences** (size, gait, movement style) → 1 and 4 were significantly different than others which is seen from the results